

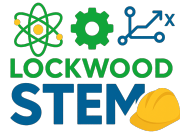
Introduction to Engineering Design

Aerospace Academy Version

Unit 5 Rubric

Introduction to Engineering Design

Course: Introduction to Engineering Design
Program: Academy of Advanced Aerospace Engineering
Brand: Lockwood STEM
Use: Student-facing and teacher grading rubrics



Unit 5

Materials, Structures & Prototype Testing

Material selection, load paths, structure design, prototype testing, failure analysis, data collection, and redesign.

Grading Rubrics and Scoring Guide

Unit 5: Materials, Structures & Prototype Testing

Assessment Purpose

Material selection, load paths, structure design, prototype testing, failure analysis, data collection, and redesign. This rubric is designed for consistent grading, feedback, and student self-checks.

Suggested 4-Point Scale

4 = Expert: exceeds expectations with clear evidence, precision, and independence. **3 = Proficient:** meets expectations with complete and accurate work. **2 = Emerging:** partially meets expectations; work is incomplete, vague, or inconsistent. **1 = Incomplete:** missing, unsafe, inaccurate, or shows limited evidence of understanding.

Unit 5 Score Summary

Category	Points Possible	Points Earned
Material selection and rationale	10	
Structural concept and load paths	15	
Prototype quality	10	
Testing procedure	15	
Data collection and analysis	15	
Failure analysis	15	
Redesign proposal	10	
Testing report/presentation	10	



Criterion	Pts	4 – Expert	3 – Proficient	2 – Emerging	1 – Incomplete
Material Selection & Rationale	10	Material choices are justified using strength, weight, manufacturability, cost, and mission needs.	Material choices are reasonable and explained.	Material explanation is vague or incomplete.	No material rationale.
Structural Concept & Load Paths	15	Design clearly shows how loads move through the structure and uses efficient lightweight geometry.	Load paths and structural choices are mostly clear.	Load paths are unclear or design is inefficient.	No clear structural concept.
Prototype Quality	10	Prototype is carefully built, follows constraints, and is appropriate for testing.	Prototype is complete and testable.	Prototype is incomplete, fragile, or does not follow constraints.	No usable prototype.
Testing Procedure	15	Procedure is repeatable, safe, includes variables/controls, and collects useful data.	Procedure is clear and mostly repeatable.	Procedure is incomplete or inconsistent.	No clear test procedure.
Data Collection & Analysis	15	Data table/graphs are complete, units are correct, and analysis connects performance to design choices.	Data and analysis are adequate.	Data are incomplete or analysis is mostly descriptive.	No useful data.
Failure Analysis	15	Explains where/why failure occurred using evidence and engineering language.	Identifies failure point and likely cause.	Failure explanation is vague or unsupported.	No failure analysis.
Redesign Proposal	10	Redesign is specific, evidence-based, and likely to improve performance.	Redesign is reasonable and connected to results.	Redesign is vague or not connected to data.	No redesign.
Testing Report / Presentation	10	Communicates design, test, data, failure, and redesign clearly and professionally.	Report/presentation is complete.	Report/presentation incomplete or hard to follow.	Missing.

Rubrics

Student Self-Check Before Submission

	Requirement	Complete?
☐	Material selection and rationale evidence is complete and ready to be graded.	
☐	Structural concept and load paths evidence is complete and ready to be graded.	
☐	Prototype quality evidence is complete and ready to be graded.	
☐	Testing procedure evidence is complete and ready to be graded.	
☐	Data collection and analysis evidence is complete and ready to be graded.	
☐	Failure analysis evidence is complete and ready to be graded.	
☐	Redesign proposal evidence is complete and ready to be graded.	
☐	Testing report/presentation evidence is complete and ready to be graded.	
☐	All pages/files are named, organized, and submitted correctly.	
☐	Reflection or revision notes are complete.	

Teacher
Use the
evidence
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